Week 8: 19/10 – 23/10

Subject: Math - Statistics **Unit Title:Unit 2: Describing and Displaying Data** **Grade: 12**

Lessons	Learning targets	Assessment tools and strategies Formative/ Summative/ Differentiation	Resources and links for Parents
Lesson 1: Stem-and-Leaf Plots and Misleading Graphs	• I can construct and interpret a stem-and-leaf plot. • I can identify and explain features that make a graph misleading.	1.2 Alternatives for auditory information (captions, transcripts, visuals). 8.3 Foster collaboration and community with structured group roles and supports. Choose an item.	Links to support learning: How to interpret a histogram | Data and statistics | 6th grade | Khan Academy Bar Charts, Pie Charts, Histograms, Stemplots, Timeplots (1.2) Measure Central Tendency : Mean-Median-Mode and Range
Lesson 2&3: Measures of centre; Choosing the fairest average; weighted and grouped means	• I can accurately compute the mean, median, mode, and midrange using the definitions and formulas provided. • I can use correct notation (μ, $\bar{x}$) and apply the rounding rule when calculating the mean, and correctly order data when finding the median. • I can explain which measure of centre best represents a data set by considering frequency, extremes, and data type.		
Lesson 4: Written Formative Assessment	Learning targets: • I can construct a categorical frequency distribution. • I can construct a grouped frequency distribution. • I can construct a histogram. • I can recognize common distribution shapes: bell-shaped, positively/right-skewed, negatively/left-skewed, J-shaped, reverse J-shaped, bimodal, and U-shaped.		

	• I can construct a stem-and-leaf plot.		
Homework	• Khan academy assigned tasks	8.1 Make goals clear, visible, and supported by short-term objectives. Choose an item.	
Teacher's notes/ Reminders	• Check **Microsoft Teams regularly** for class posts, announcements, resources, and assignments. • Keep up with your assigned **Khan Academy** skills to complete the assigned **SAT practice** and content-related skills to stay focused on preparation for both **internal assessments** and **external assessments**.		

Technology Integration	Microsoft (Power point, excel, etc.)
Cross-curricular links	**English / Media Literacy:** Students interpret survey questions, headlines, and data-based claims, then communicate and justify conclusions using clear evidence.
External Assessment Integration	SAT
Wellbeing/ ATL Skills	**Thinking Skills – Critical Thinking:** Students analyse data sources, identify bias, classify information, and evaluate whether conclusions are supported by the data. **Communication Skills:** Students discuss findings, justify choices, and use appropriate statistical vocabulary. **Research Skills:** Students plan data collection and evaluate the reliability and relevance of information.
Standards	• HS.S-ID.A.1 Model data in context with plots on the real number line (dot plots, histograms, and box plots). • HS.S-ID.A.2 Use statistics appropriate to the shape of the data distribution to compare center (median, mean) and spread (interquartile range, standard deviation) of two or more different data sets. • HS.S-ID.A.3 Interpret differences in shape, center, and spread in the context of the data sets, accounting for possible effects of extreme data points (outliers). • HS.S-IC.B.6 Evaluate reports based on data (applied here to graphical and summary reports in media and official publications).
Inquiry/ TOK Question	**How can we collect and evaluate data to make reliable conclusions about a population?**

Success Criteria (Measurable)	**Lesson 1:** • **Must:** Split values into stems and leaves, include a key, and identify a graph's axes and scale. • **Should:** Place every value in the correct row, order the leaves, read values and clusters from the plot, and explain at least two ways a graph could mislead a reader. • **Could:** Correct a misleading graph and explain how the revision changes a reader's interpretation while preserving the underlying data. **Lesson 2&3:** • **Must:** Students must accurately compute the mean, median, mode, and midrange for a given data set using the correct formulas and ordered values. • **Should:** Students should use correct statistical notation (μ, $\bar{x}$), apply the rounding rule when calculating the mean, and correctly order data when determining the median. • **Could:** Students could justify which measure of center best represents a data set by referring to frequency patterns, extreme values, and data type. **Lesson 4:** • **Must:** All students must be able to read frequency tables and data displays to identify frequencies and recognize common distribution shapes. • **Should:** Most students should be able to organize data into categorical and grouped frequency distributions and construct accurate histograms and stem-and-leaf plots. • **Could:** Some students could be able to interpret and compare data distributions, justify their conclusions using features of the displays, construct and verify a histogram.
National Identity Domain	Volunteering التطوع Global Understanding التفاهم العالمي Heritage التراث
	Area: Community Responsibility and Informed Decision-Making
	Implementation: Students community-based statistical contexts to explore how reliable data can support fair and informed decisions. They evaluate surveys, distinguish between populations and samples, classify variables, and consider how accurate data collection can help schools and UAE communities better understand needs and make responsible decisions.